Incident report analysis

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Summary	Event: Denial of Service (DoS) attack using ICMP Flooding. Cause: A Firewall was unconfigured which resulted in a massive flood of ICMP packets to enter the network. Impact: The internal network services were overwhelmed and stopped responding for two hours. This prevented staff from accessing network resources. Response: The Incident Response team blocked incoming ICMP traffic, took non critical services offline and prioritized restoring critical services.
Identify	Attack Type: Denial of Service (DoS) using ICMP flooding. Systems affected: The internal network such as the web design, graphic design, workstations, and social media marketing tools that rely on network connectivity. Vulnerability: Was the result of a misconfigured or unconfigured firewall that lacked rate limiting or ICMP filtering.
Protect	To harden the network against future floods, the security team implemented technical controls such as new firewall rules for ICMP rate limiting and an Intrusion detection system (IDS)/Intrusion Protection system (IPS) in order to

	allow for suspicious ICMP traffic to be filtered out.
Detect	The Security team implemented IP address verification on the firewall in order to detect spoofed IP addresses on incoming ICMP packets. Network Monitoring software was also implemented in order to detect any suspicious traffic.
Respond	In the case of a future event, the security team will focus on isolating the affected systems to prevent further spread/damage to the network. The team will prioritize restoring critical systems that were affected by the issue. The team will analyze network logs and look for suspicious traffic and activity. In the case anything is found, the team will communicate with upper management and the appropriate people.
Recover	In order to recover from a DoS attack by ICMP flooding, network services need to be restored to their normal state. In the future, external ICMP floods will be detected and stopped at the firewall. After, all non critical network services should be stopped to reduce internal network traffic, next critical services need to be restored and finally when the ICMP packets have timed out, all non critical network systems and services will be brought back online.
